

Atlanta Climate Change Detection: CUSUM Analysis

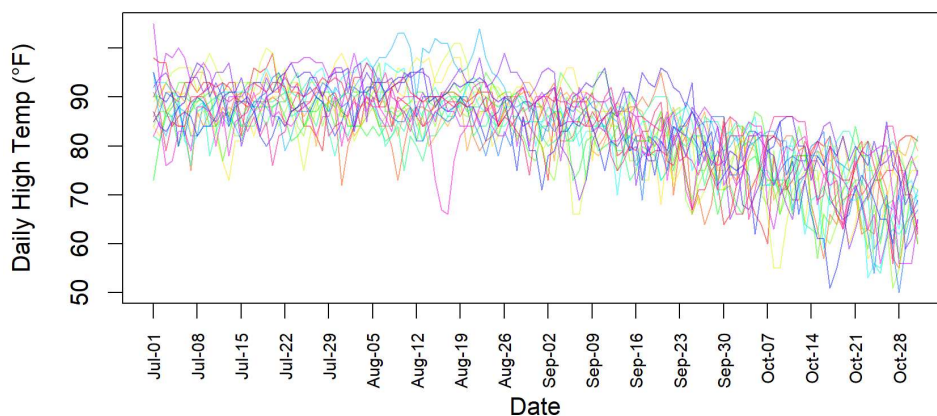
Identify “End of Summer”

Using July through October daily-high-temperature data for Atlanta for 1996 through 2015, use a CUSUM approach to identify when unofficial summer ends (i.e., when the weather starts cooling off) each year.

Methodology

1. I read in the provided temps.txt file using R and visually inspected the data, noting 123 rows and 21 columns, with the “Day” column being the date, and every other column containing the average temperature for years 1996 – 2015.
2. I then inspected the summary statistics for each year to get a sense of the overall minimum and maximum temperature values recorded for each year.
3. Next, I rendered a line plot to show the all 20 years of temperature data in order to get a sense of where to define the “mean of x, if no change” baseline period.
4. Upon visual inspection, it appears the temperatures trend downward after August, which makes empirical sense; therefore, I decided to define the baseline period for all the data up to the end of August (July 1 through August 31).

Atlanta Daily High Temperatures by Year



5. I then calculated the mean and standard deviation for each year’s temperatures during the baseline period.
6. To calculate the cumulative sum (CUSUM) using the one-sided (decreasing) formula I started with the following hyperparameters:

- a. Critical value to be 1.5 times the standard deviation ($C = 1.5 * \sigma$); and
- b. Cumulative threshold to be 5 times the critical value ($T = 5 * C$).
7. I then defined the unofficial last day of summer for each year to be the date associated with when the threshold was reached.
8. I then reviewed plots for each year to see where the threshold was reached to see if it appeared reasonably near where temperatures consistently dropped.
9. I returned the hyperparameters (C and T) and repeated the process multiple times until the threshold appeared to consistently be reached close to when temperatures consistently fell below the mean.

Results

Method: One-sided CUSUM detecting sustained drops below the baseline mean, with

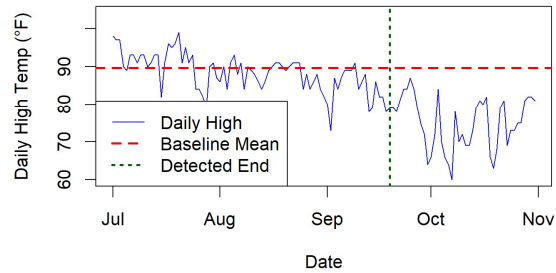
- $C = 1 \times \sigma$ (standard deviation of baseline through 8/31)
- $T = 7 \times C$ (7-day cumulative threshold)

Here are the unofficial end of summer dates that result:

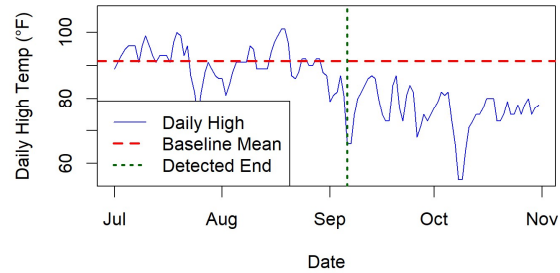
Year	End of Summer Date	Temperature
1996	19-Sep	79
1997	26-Sep	70
1998	29-Sep	75
1999	22-Sep	72
2000	6-Sep	66
2001	3-Sep	73
2002	24-Sep	73
2003	10-Sep	79
2004	16-Sep	75
2005	8-Oct	73

Year	End of Summer Date	Temperature
2006	13-Sep	70
2007	30-Sep	79
2008	19-Sep	81
2009	2-Sep	78
2010	28-Sep	76
2011	6-Sep	75
2012	21-Sep	85
2013	15-Oct	68
2014	25-Sep	77
2015	15-Sep	80

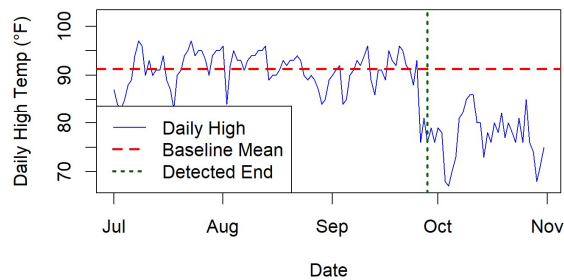
Daily High Temps & End-of-Summer for X1996



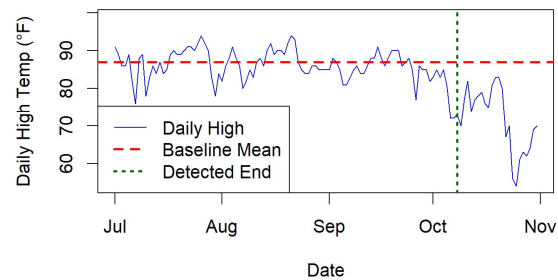
Daily High Temps & End-of-Summer for X2000



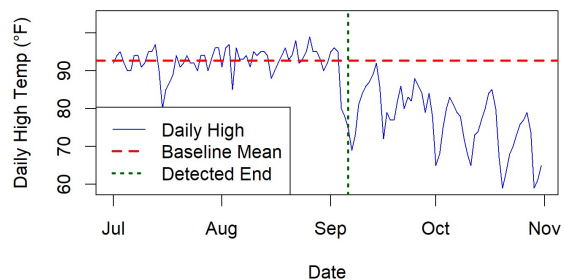
Daily High Temps & End-of-Summer for X2010



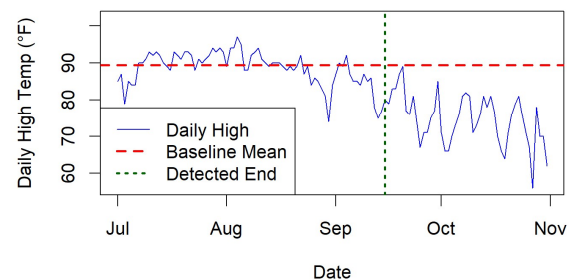
Daily High Temps & End-of-Summer for X2005



Daily High Temps & End-of-Summer for X2011



Daily High Temps & End-of-Summer for X2015



Discussion of Results

- **Observed Range:**
 - Earliest detection: **September 3 (2001)**
 - Latest detection: **October 15 (2013)**
 - Majority of years fall in **early-to-late September**, shortly after the summer plateau ends.
- **Notable Outliers:**
 - **October detections:** 2005 (Oct 8), 2013 (Oct 15) — unusually late relative to typical seasonal decline

- **Early detections:** 2000 & 2001 (Sept 3–6) — may reflect unusually early post-peak cooling

Has Summer Gotten Warmer (between 1996 – 2015)?

Use a CUSUM approach to make a judgment of whether Atlanta’s summer climate has gotten warmer in that time (and if so, when).

Methodology

1. I defined “summer” for this exercise to be Jul 1 through August 31, given they are the hottest.
2. I then calculated the summer mean temperature for each year.
3. For the baseline summer period, I selected the first 3 years of the data set (1996, 1997, and 1998).
4. I then calculated the mean and standard deviation for those three years.
5. I then tried different combinations of C and T hyperparameters calculated as multiples of the standard deviation ($C=[.5, 1, 1.5]$ and $T=[2, 2.5, 3]$) with CUSUM to see if there was some consistency around the year(s) where the threshold was reached.
6. After selecting my final C and T hyperparameters, I then plotted the results.

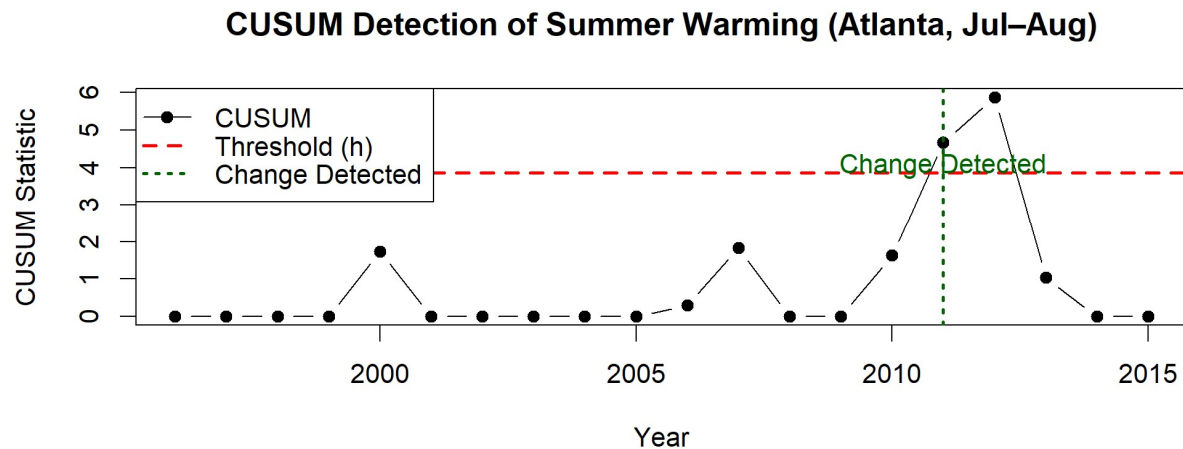
Results & Discussion

Baseline mean (μ): 88.12903

Baseline SD (σ): 1.543423

Formula	First_Crossing_Year	Max_CUSUM
$C = 0.772, T = 3.087$	2000	8.581891
$C = 1.543, T = 3.087$	2011	5.869732
$C = 2.315, T = 3.087$	2011	3.554597
$C = 0.772, T = 3.859$	2011	8.581891
$C = 1.543, T = 3.859$	2011	5.869732
$C = 2.315, T = 3.859$	NA	3.554597
$C = 0.772, T = 4.63$	2011	8.581891
$C = 1.543, T = 4.63$	2011	5.869732
$C = 2.315, T = 4.63$	NA	3.554597

Across the various hyperparameters tried, the threshold seemed to hold steady on the same year (2011), so I opted for $C=(1 \text{ sigma})$ and $T=(2.5 * \text{sigma})$.



Across the 9 combinations of C and T values tried, the threshold seemed to trigger for 2011 or not at all. I ultimately decided to use the reference/critical value to be 1 standard deviation and the cumulative threshold to be 2.5 standard deviations, based on the baseline variability of mean July–August temperatures. The goal for this threshold was to avoid triggering on typical year-to-year fluctuations, while still being sensitive enough to detect a meaningful upward shift.

The results show that the CUSUM statistic exceeded the threshold only in 2011. However, this spike was **isolated**, and the CUSUM quickly returned below the threshold in subsequent years, indicating that it was a single anomalously warm year rather than a sustained increase in summer temperatures. Therefore, while the chosen parameters successfully filtered out minor variability, they also highlight that **no persistent warming trend is evident in this dataset**.

References

The methodology I used in R was informed by:

- <https://www.geeksforgeeks.org/r-language/how-to-perform-grubbs-test-in-r/>
- https://en.wikipedia.org/wiki/Grubbs%27s_test
- https://en.wikipedia.org/wiki/Freedman%E2%80%93Diaconis_rule
- <https://www.itl.nist.gov/div898/handbook/pmc/section3/pmc323.htm>
- <https://sixsigmastudyguide.com/cumulative-sum-chart-cusum/>